

# Applications



1

a 13%

b \$56 500

2

a  $\frac{4}{13}$

b \$1650

c Option A

3

a  $\frac{11}{100}$

b  $\frac{6}{55}$

c

d \$18.90

|            |     |      |      |
|------------|-----|------|------|
| $x$        | 12  | 15   | 25   |
| $P(X = x)$ | 0.2 | 0.35 | 0.45 |

e \$5.62

4

a  $\frac{27}{64}$

b  $\frac{1}{64}$

c  $\frac{27}{64}$

d

|            |                 |                 |                |                |
|------------|-----------------|-----------------|----------------|----------------|
| $x$        | 0               | 1               | 2              | 3              |
| $P(X = x)$ | $\frac{27}{64}$ | $\frac{27}{64}$ | $\frac{9}{64}$ | $\frac{1}{64}$ |

e Yes, because  $\sum P(x) = 1$ ,  $X$  is discrete numerical, and  $0 \leq P(x) \leq 1$ .

5

a

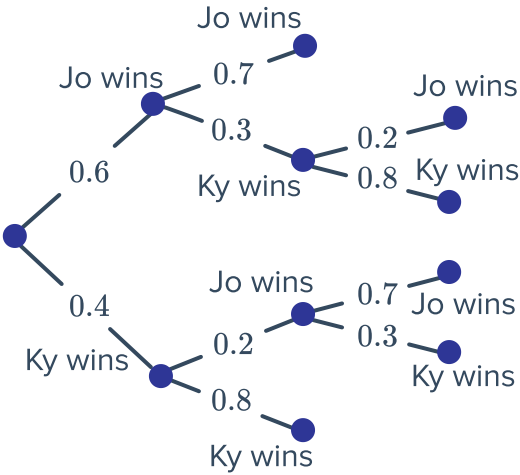
|            |     |     |     |     |     |
|------------|-----|-----|-----|-----|-----|
| $x$        | 0   | 1   | 2   | 3   | 4   |
| $P(X = x)$ | 0.4 | 0.1 | 0.2 | 0.2 | 0.1 |

b 1.5

c  $\frac{1}{5}$

6

a



b



| $x$        | 2    | 3    |
|------------|------|------|
| $P(X = x)$ | 0.74 | 0.26 |

c 2.26

d  $\frac{23}{128}$

7

a

| $x$        | 1             | 2             | 3             | 4             | 5             | 6             |
|------------|---------------|---------------|---------------|---------------|---------------|---------------|
| $P(X = x)$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ |

b Uniform

c 3.5

d  $\frac{35}{12}$

8

a

| $y$        | 2              | 3              | 4              | 5              | 6              | 7              | 8              | 9              | 10             | 11             | 12             |
|------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| $P(Y = y)$ | $\frac{1}{36}$ | $\frac{2}{36}$ | $\frac{3}{36}$ | $\frac{4}{36}$ | $\frac{5}{36}$ | $\frac{6}{36}$ | $\frac{5}{36}$ | $\frac{4}{36}$ | $\frac{3}{36}$ | $\frac{2}{36}$ | $\frac{1}{36}$ |

b 7

c The distribution is symmetrical.

d 7

e  $\frac{35}{6}$

9



a

|            |               |               |               |               |               |               |
|------------|---------------|---------------|---------------|---------------|---------------|---------------|
| $x$        | 1             | 2             | 3             | 4             | 5             | 6             |
| $P(X = x)$ | $\frac{2}{9}$ | $\frac{1}{9}$ | $\frac{2}{9}$ | $\frac{1}{9}$ | $\frac{2}{9}$ | $\frac{1}{9}$ |

b  $\frac{10}{3}$

c  $\frac{26}{9}$

10

a

|            |               |                |                |                |                |
|------------|---------------|----------------|----------------|----------------|----------------|
| $x$        | 2             | 3              | 4              | 5              | 6              |
| $P(X = x)$ | $\frac{1}{6}$ | $\frac{5}{24}$ | $\frac{5}{24}$ | $\frac{5}{24}$ | $\frac{5}{24}$ |

b  $\frac{49}{12}$

c  $\frac{275}{144}$

11

a

|            |     |     |     |     |     |     |
|------------|-----|-----|-----|-----|-----|-----|
| $x$        | 1   | 2   | 3   | 4   | 5   | 6   |
| $P(X = x)$ | 0.1 | 0.1 | 0.1 | 0.3 | 0.1 | 0.3 |

b

i 0.2

ii  $\frac{1}{7}$

iii  $\frac{1}{3}$

iv  $\frac{2}{9}$

12

a

|            |               |               |
|------------|---------------|---------------|
| $y$        | 2             | 3             |
| $P(Y = y)$ | $\frac{1}{2}$ | $\frac{1}{2}$ |

b Uniform

13

a

|            |               |               |               |               |               |               |
|------------|---------------|---------------|---------------|---------------|---------------|---------------|
| $w$        | 15            | 16            | 17            | 18            | 19            | 20            |
| $P(W = w)$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ | $\frac{1}{6}$ |

b Uniform

14

a 0, 4, 8



b

| $x$        | 0             | 4             | 8             |
|------------|---------------|---------------|---------------|
| $P(X = x)$ | $\frac{1}{6}$ | $\frac{4}{6}$ | $\frac{1}{6}$ |

15

a

|   | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 2 | 3 | 4 | 5 |
| 2 | 1 | 0 | 1 | 2 | 3 | 4 |
| 3 | 2 | 1 | 0 | 1 | 2 | 3 |
| 4 | 3 | 2 | 1 | 0 | 1 | 2 |
| 5 | 4 | 3 | 2 | 1 | 0 | 1 |
| 6 | 5 | 4 | 3 | 2 | 1 | 0 |

b

| $x$        | 0              | 1               | 2              | 3              | 4              | 5              |
|------------|----------------|-----------------|----------------|----------------|----------------|----------------|
| $P(X = x)$ | $\frac{6}{36}$ | $\frac{10}{36}$ | $\frac{8}{36}$ | $\frac{6}{36}$ | $\frac{4}{36}$ | $\frac{2}{36}$ |

c  $\frac{2}{3}$

d  $\frac{9}{10}$

16

a

|   | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 2 | 3 | 4 | 5 |
| 2 | 1 | 0 | 1 | 2 | 3 | 4 |
| 3 | 2 | 1 | 0 | 1 | 2 | 3 |
| 4 | 3 | 2 | 1 | 0 | 1 | 2 |
| 5 | 4 | 3 | 2 | 1 | 0 | 1 |
| 6 | 5 | 4 | 3 | 2 | 1 | 0 |

b

| $x$        | \$0             | \$1            | \$2            | \$3            |
|------------|-----------------|----------------|----------------|----------------|
| $P(X = x)$ | $\frac{24}{36}$ | $\frac{6}{36}$ | $\frac{4}{36}$ | $\frac{2}{36}$ |

c \$0.56

d -\$1.44

17

a

|            |               |               |               |
|------------|---------------|---------------|---------------|
| $x$        | \$3           | \$6           | \$9           |
| $P(X = x)$ | $\frac{3}{6}$ | $\frac{2}{6}$ | $\frac{1}{6}$ |



b \$5

c \$2.24

18

a 0.76

b \$1.69

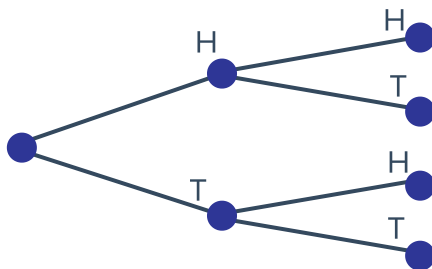
c  $C = \$1.50$

19

|            |      |      |      |
|------------|------|------|------|
| $x$        | 0    | 1    | 2    |
| $P(X = x)$ | 0.09 | 0.42 | 0.49 |

20

a



b

|            |               |               |               |
|------------|---------------|---------------|---------------|
| $x$        | -\$3          | \$0           | \$3           |
| $P(X = x)$ | $\frac{1}{4}$ | $\frac{1}{2}$ | $\frac{1}{4}$ |

21

a 0.147

b 0.1029

c 0.07203

d  $P(N = n) = (0.7)^n \times 0.3$  for  $n = 0, 1, 2, 3, \dots$

22

a 1, 2, 3, 4, 5, 6, 7, 8

b

|            |                |                |                |                |                                                                                  |                |                |                |
|------------|----------------|----------------|----------------|----------------|----------------------------------------------------------------------------------|----------------|----------------|----------------|
| $x$        | 1              | 2              | 3              | 4              |  | 6              | 7              | 8              |
| $P(X = x)$ | $\frac{3}{16}$ | $\frac{3}{16}$ | $\frac{2}{16}$ | $\frac{2}{16}$ | $\frac{2}{16}$                                                                   | $\frac{2}{16}$ | $\frac{1}{16}$ | $\frac{1}{16}$ |

23

a

|            |               |               |               |               |
|------------|---------------|---------------|---------------|---------------|
| $x$        | 1             | 2             | 3             | 4             |
| $P(X = x)$ | $\frac{1}{2}$ | $\frac{1}{4}$ | $\frac{1}{8}$ | $\frac{1}{8}$ |

b

|            |               |               |               |   |
|------------|---------------|---------------|---------------|---|
| $x$        | 1             | 2             | 3             | 4 |
| $P(X = x)$ | $\frac{1}{2}$ | $\frac{3}{4}$ | $\frac{7}{8}$ | 1 |

c

- i  $\frac{3}{4}$     ii  $\frac{1}{4}$     iii  $\frac{5}{8}$     iv  $\frac{3}{4}$

24

a 0, 1, 2, 3, 4, 6, 8, 9, 12, 16

b

|            |                |                |                |                |                |                |                |                |                |                |
|------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| $x$        | 0              | 1              | 2              | 3              | 4              | 6              | 8              | 9              | 12             | 16             |
| $P(X = x)$ | $\frac{9}{25}$ | $\frac{1}{25}$ | $\frac{2}{25}$ | $\frac{2}{25}$ | $\frac{3}{25}$ | $\frac{2}{25}$ | $\frac{2}{25}$ | $\frac{1}{25}$ | $\frac{2}{25}$ | $\frac{1}{25}$ |

c 4

d 20

25

a

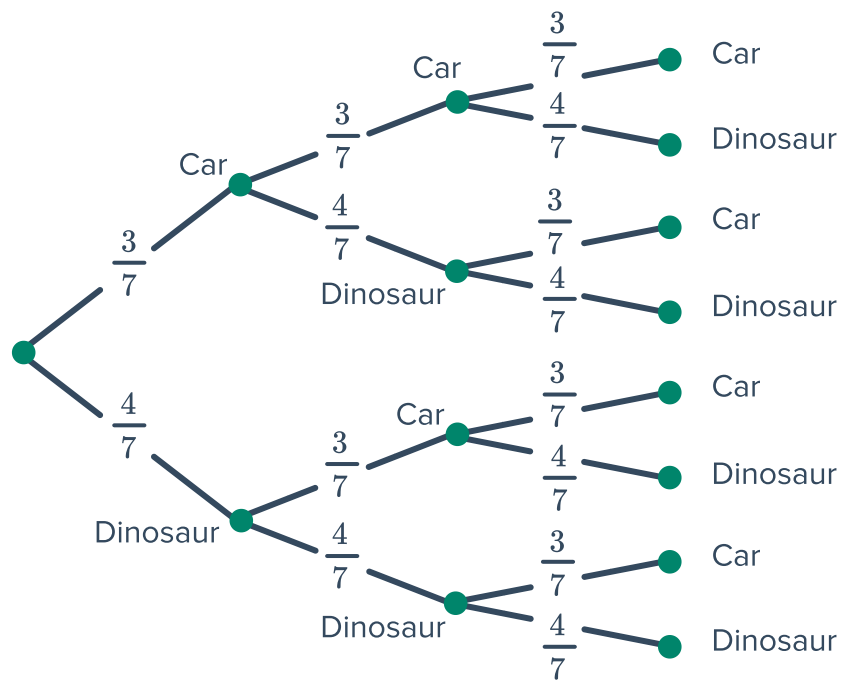
|            |                   |                    |                    |                    |
|------------|-------------------|--------------------|--------------------|--------------------|
| $x$        | 0                 | 1                  | 2                  | 3                  |
| $P(X = x)$ | $\frac{27}{1000}$ | $\frac{189}{1000}$ | $\frac{441}{1000}$ | $\frac{343}{1000}$ |

b

|            |                 |                   |                   |                   |
|------------|-----------------|-------------------|-------------------|-------------------|
| $x$        | 0               | 1                 | 2                 | 3                 |
| $P(X = x)$ | $\frac{6}{720}$ | $\frac{126}{720}$ | $\frac{378}{720}$ | $\frac{210}{720}$ |

26

a



b

| $x$        | 0                | 1                 | 2                 | 3                |
|------------|------------------|-------------------|-------------------|------------------|
| $P(X = x)$ | $\frac{64}{343}$ | $\frac{144}{343}$ | $\frac{108}{343}$ | $\frac{27}{343}$ |

c  $1\frac{2}{7}$  daysd  $\frac{6}{7}$  days

27

a  $\frac{2}{11}$ b  $\frac{10}{33}$ 

c

| $X$        | 0              | 1              | 2              | 3               | 4              |
|------------|----------------|----------------|----------------|-----------------|----------------|
| $P(X = x)$ | $\frac{1}{66}$ | $\frac{2}{11}$ | $\frac{5}{11}$ | $\frac{10}{33}$ | $\frac{1}{22}$ |

28

a  $\frac{1125}{9604}$ b  $\frac{3645}{9604}$

c

| $x$        | 0                     | 1                   | 2                      |  | 3                   | 4                      |
|------------|-----------------------|---------------------|------------------------|----------------------------------------------------------------------------------|---------------------|------------------------|
| $P(X = x)$ | $\frac{625}{38\,416}$ | $\frac{1125}{9604}$ | $\frac{6075}{19\,208}$ |                                                                                  | $\frac{3645}{9604}$ | $\frac{6561}{38\,416}$ |

29

a

|            |                |                 |                |
|------------|----------------|-----------------|----------------|
| $x$        | 0              | 1               | 2              |
| $P(X = x)$ | $\frac{5}{14}$ | $\frac{15}{28}$ | $\frac{3}{28}$ |

b  $\frac{1}{6}$

30

a

|            |                |                |                |
|------------|----------------|----------------|----------------|
| $x$        | 0              | 1              | 2              |
| $P(X = x)$ | $\frac{7}{15}$ | $\frac{7}{15}$ | $\frac{1}{15}$ |

b

|            |                |                |                |
|------------|----------------|----------------|----------------|
| $t$        | 3              | 7              | 11             |
| $P(T = t)$ | $\frac{7}{15}$ | $\frac{7}{15}$ | $\frac{1}{15}$ |

c 3, 7

d 5.4